

Skills Summary

Reading a Line Plot with Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading counts off the plot

Rule: On a line plot, the *tick* tells you the amount and the *height of the stack* tells you how many.

So “how many” questions count crosses, and “how much” questions use the fraction under them. Reading the wrong one is the mistake this plot invites.

Number of measurements = the sum of all the stack heights.

Example: A plot has 3 crosses at $\frac{1}{4}$, 5 crosses at $\frac{1}{2}$ and 2 crosses at $\frac{3}{4}$. How many measurements were taken?

Step 1. The question asks how many, not how much, so I add stack heights and ignore the fractions underneath.

Step 2. $3 + 5 + 2 = 10$ crosses, and each cross is one measurement. Measurements taken: **10**

Try it: A plot has 2 crosses at $\frac{1}{8}$, 4 at $\frac{3}{8}$ and 3 at $\frac{5}{8}$. How many measurements were taken?

check: 6

Skill 2 — Adding up one whole stack

Rule: A stack of n crosses at amount a holds $n \times a$ altogether.

Multiply the numerator by n and leave the denominator alone: $n \times \frac{p}{q} = \frac{n \times p}{q}$. Then simplify, and write anything over 1 as a mixed number.

Example: Three containers each hold $\frac{1}{4}$ of a litre. How much is that altogether?

Step 1. Equal amounts repeated is multiplication, so I multiply the amount by the number of containers rather than adding three times.

Step 2. $3 \times \frac{1}{4} = \frac{3}{4}$, which is already in lowest terms and less than 1. Total: $\frac{3}{4}$ of a litre

Try it: Five containers each hold $\frac{3}{8}$ of a litre. How much is that altogether?

check: $\frac{15}{8}$ litres

Skill 3 — Finding the middle of the data

Rule: The **median** is the value in the middle place once the data is in order — and a line plot is already in order, left to right.

With an odd number of crosses, count in to the middle one. With an even number, the median sits halfway between the two middle crosses. The **range** is the rightmost amount minus the leftmost.

Example: Seven measurements: $\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{3}{8}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}$. Find the median.

Step 1. The list is already ordered, so the median is a counting job: find the middle place rather than doing any arithmetic.

Step 2. Seven values means the middle place is the 4th, and counting in from the left the 4th value is $\frac{3}{8}$. Median: $\frac{3}{8}$

Try it: Seven measurements: $\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \frac{5}{8}, \frac{3}{4}, \frac{3}{4}$. Find the median.

check: $\frac{3}{4}$

Skill 4 — Pouring together and sharing out

Rule: Two steps, in this order. Pour together: $\text{total} = n \times a$. Share equally: $\text{each share} = \text{total} \div m$.

Dividing a fraction by a whole number means multiplying by its reciprocal: $\frac{p}{q} \div m = \frac{p}{q \times m}$. Doing the sharing before the pouring gives the wrong answer.

Example: Six cups each hold $\frac{1}{2}$ litre. They are poured together, then shared equally between 4 cups. How much is in each?

Step 1. Sharing only works on one pooled amount, so I total the water first and divide afterwards.

Step 2. Pour: $6 \times \frac{1}{2} = 3$ litres. Share: $3 \div 4 = \frac{3}{4}$. Each cup: $\frac{3}{4}$ of a litre

Try it: Four cups each hold $\frac{3}{4}$ litre. They are poured together, then shared equally between 6 cups. How much is in each?

check: $\frac{3}{4}$ of a litre